

26t1_NPPE1 Report

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Introduction

Problem statement:

1. In clinical field missing a disease or diagnosis (False Negative) is significantly more dangerous than a false alarm. By using pretrained deep learning models we can train a model with the given X-ray dataset to early and accurate detection of thoracic diseases.
2. Dataset: The dataset consist of 51,043 chest X-ray images categorised into 20 distinct classes.
3. Objective: To make precise detection model should penalizes False Negative (FN) five times more heavily than False Positive (FP)

Methodology

1. Models used:
 - ResNet50: Chosen for its depnth and ability to learn complex features through residual connections.
 - ResNet34: Later adopted as a lightweight alternative to ensure stable running within the memory limits of the T4 GPU.
2. Preprocessing:
 - Resizing: All the images were standardized to 224*224 pixels to balance detail with processing speed.
 - Normalization applied using ImageNet mean and standard deviation.
3. Key Techniques:
 - Transfer Learning: Initialized models with IMAGENET1K_V1 weights to leverage pre-learned visual features.
 - Log smoothed class weights: Implemented a custom weighting strategy ($1 / \log(1 + \text{count})$) to prevent the model from being biased toward the majority "No Finding" class.
 - Gradient Accumulation: Used to simulate an effective batch size of 32 while maintaining a small physical batch size of 8, preventing Out-of-Memory (OOM) errors in kaggle environment.

- One-Hot Encoding: Enforced a submission format where exactly one pathology is predicted per image

Experiments:

1. Model Comparison: Evaluated the performance shift from the heavier ResNet50 to the more efficient ResNet34 .
2. Parameter Changes:
 - Batch Size: Scaled down from 64 (Dual GPU) to 8 (Single GPU stability) to accommodate hardware constraints.
 - Optimizer: Transitioned to AdamW for better weight decay handling and more stable convergence.
3. Validation Strategy: Train and Validation Split: Employed a 12% hold-out validation set (88 & 12 split) to monitor generalization and competition score during training.

Results:

Model	Epoch	Best Training Loss	Best Validation Loss	Kaggle Score
ResNet50	15	2.5287	2.3435	-15.1172
ResNet34	30	1.0500	1.0525	-11.4565

Conclusion:

The project demonstrated that architectural stability and class-balance management are more critical than model depth in medical imaging tasks. By switching to a more sustainable ResNet34 and doubling the training duration, the model was able to converge more effectively.

Key Learnings: GPU of the environment can be handled productively with the right Neural Network architecture and batch size.